



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - JAN/FEB 2023

SUBJECT : ECE 2151 - ANALOG ELECTRONIC CIRCUITS
(MAKEUP)

ANALOG ELECTRONIC CIRCUITS [ECE 2151]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) The circuit shown in FIG.Q1(A) must be designed for a voltage drop of 250 mV across R_S . Given $V_{DD} = 1.5V$, $R_D = 500\Omega$, $R_S = 150\Omega$, calculate the minimum allowable value of W/L if M_1 must remain in saturation. Assume $\lambda = 0$, $V_{TH} = 0.5V$, $\mu_n C_{ox} = 100\mu A/V^2$.

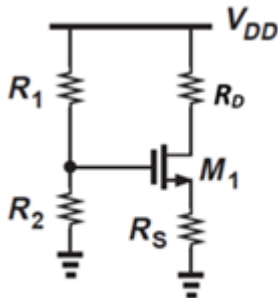


FIG.Q1(A)

- 1B) Identify the circuit shown in FIG.Q1(B). find the voltage gain of this circuit for $\lambda \neq 0$.

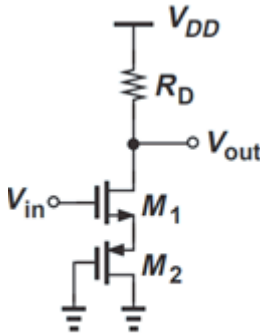


FIG.Q1(B)

- 1C) A MOSFET carries a drain current of 1 mA with $V_{DS} = 0.5 V$ in saturation. Determine the change in I_D if V_{DS} rises to 1 V and $\lambda = 0.1$. What is the device output impedance? (3)
- 2A) Determine the V_{out2}/V_{out1} for the given circuit shown in FIG.Q2(A) (Assume $\lambda=0$ for M_1 and $\lambda \neq 0$ for rest of the MOSFETs) (4)

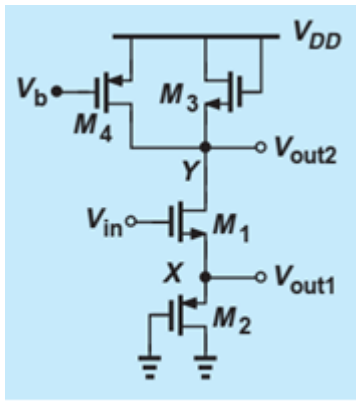


FIG.Q2(A)

- 2B) For the Cascode amplifier shown in FIG.Q2(B), $(W/L)_1 = 30$, $(W/L)_2 = 40$, $I_1 = 0.5 \text{ mA}$. If $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $\lambda_n = 0.1 \text{ V}^{-1}$, find the voltage gain.

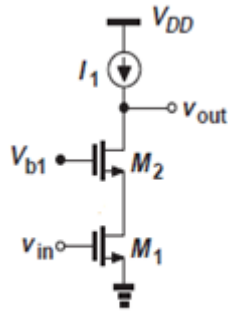


FIG.Q2(B)

- 2C) Determine 'Icopy' for the circuit shown in FIG.Q2(C)

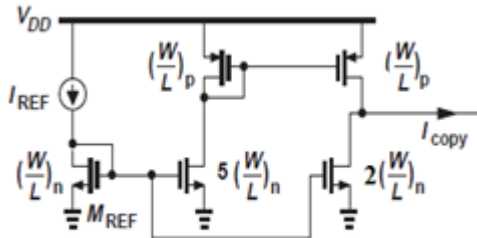


FIG.Q2(C)

- 3A) Find the differential gain, common mode gain and CMRR of the circuit shown in FIG.Q3(A). Assume $\lambda \neq 0$ and ideal current source 'ISS'.

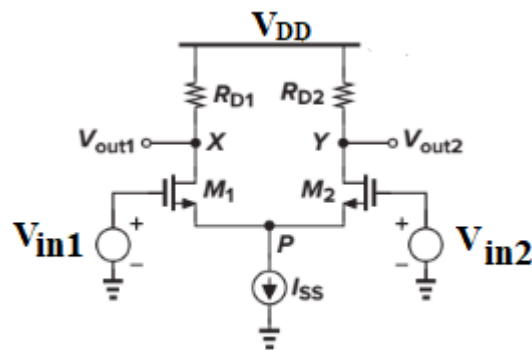


FIG.Q3(A)

- 3B) For the circuit shown in FIG.Q3(B), find the poles at low frequencies. Also, sketch its low frequency response. Assume $\lambda = 0$.

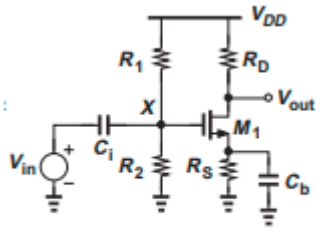


FIG.Q3(B)

- 3C) For the circuit shown in FIG.Q3(C), sketch the frequency response. Assume $\lambda = 0$, $g_m = 0.003\Omega^{-1}$, $C_{GD} = 80\text{fF}$ and $C_{SB} = 100\text{fF}$.

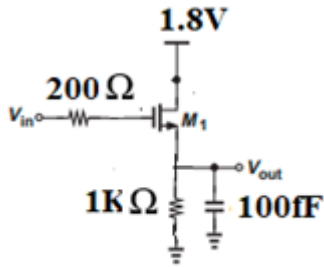


FIG.Q3(C)

- 4A) Determine the closed loop gain, I/O impedances of the circuit shown in FIG. Q4(A).

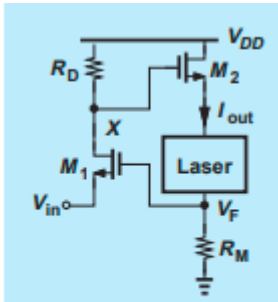


FIG.Q4(A)

- 4B) For the block diagram shown in FIG. Q4(B), Identify the feedback topology and obtain the expressions for I/O impedances.

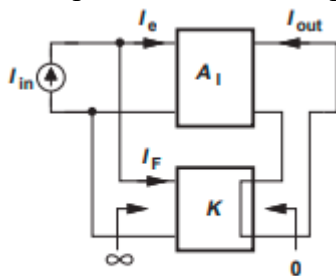


FIG.Q4(B)

- 4C) Determine the polarity of feedback for the circuit shown in FIG.Q4(C) with justification.

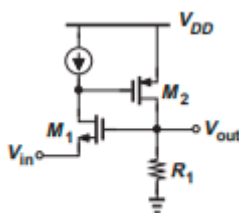


FIG.Q4(C)

- 5A) Explain the working principle of RC phase shift Oscillator with a neat circuit diagram. Obtain the expressions for minimum gain and frequency of oscillation. Also, find the frequency of oscillation for $R=1K\Omega$ and $C=1\mu F$. (4)
- 5B) Derive an expression for efficiency of Class -B Push -Pull Power amplifier. (3)
- 5C) Explain the working principle of ring oscillator with a neat circuit diagram. Obtain the expressions for minimum gain and frequency of oscillation. Also, find the frequency of oscillation for drain resistance and capacitance of $1K\Omega$ and $1\mu F$. (3)

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